Place Value & the Hindu System

The modern system uses just ten symbols, a place value for each position, and a zero. Learn how place value and expanded form work, why zero matters, and the base-60 legacy we still use.

Place value & expanded form

The idea: In the Hindu (decimal) system the value of a digit depends on its **place**. Moving one place to the left multiplies the place value by 10.

Place	Ten-thousands	Thousands	Hundreds	Tens	Ones
Value	10^4	10^3	10^2	10^1	10^0
Digit (47561)	4	7	5	6	1

$$47561 = (4 \times 10^4) + (7 \times 10^3) + (5 \times 10^2) + (6 \times 10^1) + (1 \times 10^0)$$

Face value vs place value

- **Face value** = the digit itself (the 6 in 36174 has face value 6).
- **Place value** = digit × its place (the 6 in 36174 is in the thousands place, so place value 6000).

The role of zero

Zero is a **placeholder**. It keeps the other digits in the right places, so 52, 502 and 520 are all different numbers. A system without zero (like Roman or Egyptian) cannot do this cleanly.

The base-60 legacy

The Mesopotamian system became **base 60** (sexagesimal). We still use base 60 today: 60 seconds = 1 minute, 60 minutes = 1 hour, and 360° in a full circle.

Why the Hindu system won: only ten symbols (0–9), a place value for each position, and a symbol for zero. Together these make both writing numbers and doing arithmetic simple — which is why the whole world uses it.

Prerequisite refresher: "Expanded form" writes a number as the sum of each digit times its place value.

Where marks slip away

Trap 1 — Face value ≠ place value. The 6 in 36174 has face value 6 but place value 6000.

Trap 2 — Count the places carefully. Ones = 10^0, tens = 10^1, hundreds = 10^2, … Read right to left.

Trap 3 — Don't drop the zeros. In expanded form a zero digit contributes 0, but it still holds the place ($5092 = 5\times1000 + 0\times100 + 9\times10 + 2$).

Trap 4 — Base 60 ≠ base 10. In time and angles the carrying happens at 60, not 10.

Slip 5 — Write powers of 10 clearly (10^3 not 103) in expanded form.

Model answers — write them like this

Example A. Write 47561 in expanded form using powers of 10.

Step 1: Give each digit its place value.

Step 2: $47561 = (4 \times 10^4) + (7 \times 10^3) + (5 \times 10^2) + (6 \times 10^1) + (1 \times 10^0)$.

∴ $(4 \times 10^4) + (7 \times 10^3) + (5 \times 10^2) + (6 \times 10^1) + (1 \times 10^0)$

Example B. Find the place value of the digit 8 in 38420.

Step 1: The 8 is in the thousands place (10^3).

Step 2: Place value = $8 \times 1000 = 8000$.

∴ 8000

Example C. Find the place value and face value of the digit 6 in 36174, and their difference.

Step 1: Face value = 6. The 6 is in the thousands place, so place value = 6000.

Step 2: Difference = $6000 - 6 = 5994$.

∴ Place 6000, face 6, difference 5994

Example D. Why is zero important in the place-value system?

Step 1: Zero holds an empty place so the other digits stay in position.

Step 2: Without it, 52, 502 and 520 could not be told apart.

∴ Zero is a placeholder that keeps each digit's place value correct

Next: try the worksheet, then check the answer key.